APPLE DEVELOPER ACADEMY INDONESIA — TECH TRACK APPLICANT

Sumit Saraswat

AI Engineer & Healthcare Innovator

"The most powerful technology is the technology people trust enough to use."

Showcasing privacy-first healthcare AI, edge computing, equity-aware medical modeling, and my journey from a solo freshman to the top 100 at Meta's nationwide hackathon. I build AI that respects human dignity.

CORE TECH ARSENAL

Machine Learning & Data: Python, PyTorch (RL Frameworks), SEER Registry Auditing

Edge & Web Integration: JavaScript (Vanilla Edge NLP), WebSpeech API, HTML/CSS

Design & Ecosystem: Figma, Apple HIG Concept Prototyping

▶ VIEW INTERACTIVE WEB PORTFOLIO

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1. Meta OpenEnv Hackathon

RL Auditor & SynthAudit Benchmark (Top 100 Finals).

TYPE	ROLE
Self-Initiated Competitive Sprint (Top 100 Finals)	**Solo Competitor**

ROUND 1: THE CLINICAL AUDITOR & BENCHMARK

I entered this nationwide competition as a solo freshman. I developed an autonomous RL clinical auditing environment and the SynthAudit benchmark. **Meta's world's strongest frontier model, Llama 3.1 405B, scored only 0.34 out of 1.0 (just 34%)** navigating the highly complex autonomous environment I built. I made it to the **Top 800 in Round 1 out of 32,000+ teams** of senior developers and working professionals.

ROUND 2: THE FINAL BANGALORE SHOWDOWN

In the offline finals in Bangalore, I competed against the 800 best senior developer teams and working professionals. I officially made it to the **Top 100** nationwide.

WHAT I LEARNED & ACHIEVED

Clarity of thought beats sheer manpower. I gained deep expertise in Python, RL, and Docker. Furthermore, **I trained the AI model and the overall performance increased by 283%.**

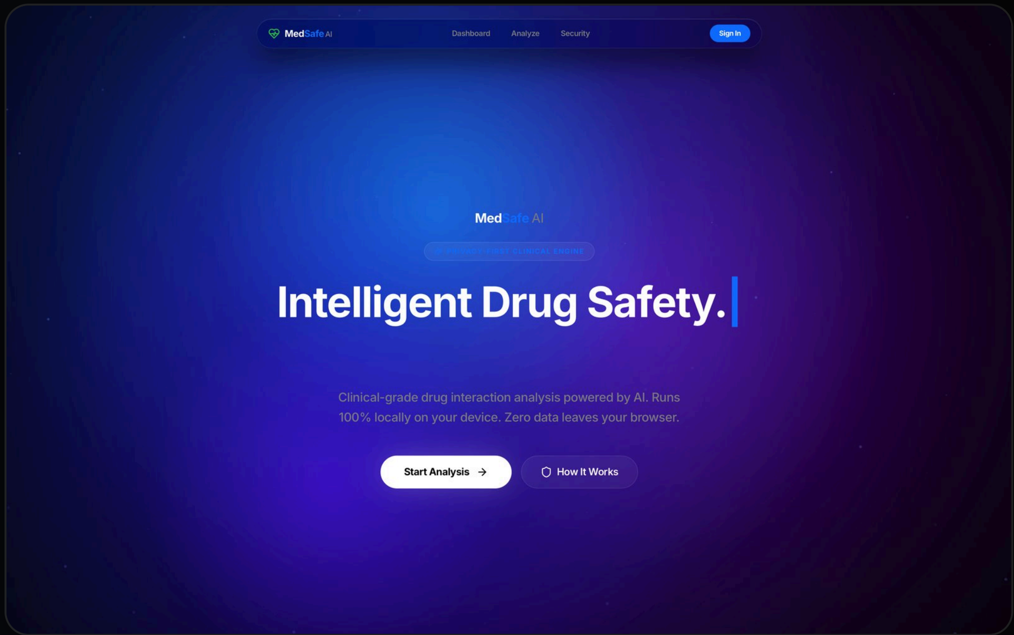
Round 1 Repo

Round 2 Repo

2. MedSafe AI

Privacy-preserving on-device clinical pharmacologist.

TYPE	ROLE
Hackathon Project	Solo Full-Stack AI Engineer



ONE-SENTENCE SUMMARY

An offline-first, WebGPU-accelerated local LLM (Llama 3.2 1B) that analyzes complex drug-food-supplement interactions using on-device OCR, ensuring zero latency and 100% HIPAA-compliant data privacy.

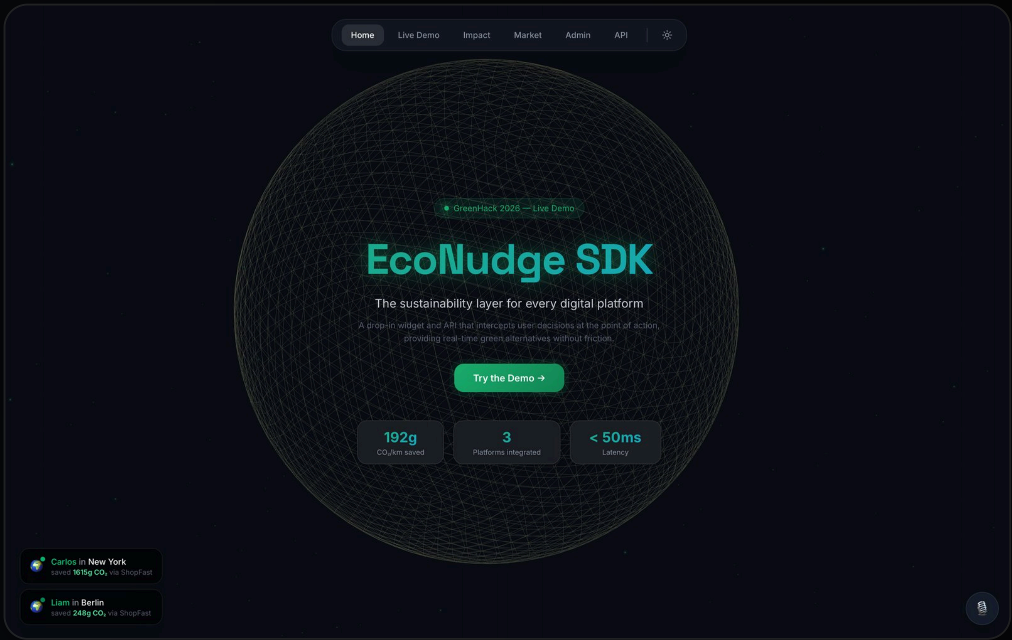
THE IMPACT

Polypharmacy leads to millions of adverse drug events annually. Current AI solutions send highly sensitive patient data to remote cloud servers, causing severe privacy violations. I architected MedSafe AI to run a 1-Billion parameter Llama 3.2 model entirely in the browser via WebGPU. By integrating local OCR prescription scanning and the Web Speech API, this tool acts as a private, on-device clinical pharmacologist, keeping patient data strictly on their own hardware.

WHAT I LEARNED

I learned how to push the boundaries of Edge Computing and Local AI. Compiling a Large Language Model to run inside a standard web browser taught me WebGPU memory constraints and hardware-accelerated inferencing. Designing the Apple Siri-inspired dynamic glassmorphism UI reinforced my belief that advanced AI must be accessible, responsive, and visually intuitive.

Live Website ↗ | YouTube ▶ | GitHub



3. EcoNudge SDK — The Sustainability Layer for the Internet

TYPE	ROLE
Team Project	Lead Architect & Spatial Engineer

ONE-SENTENCE SUMMARY

An invisible, AI-driven SDK that injects carbon-intelligence into existing e-commerce flows, nudging users toward sustainable choices through 3D gamification and real-time predictive modeling.

THE IMPACT

We solved the "Sustainability Friction" problem. By engineering a headless React SDK with sub-50ms latency, we created a way for enterprises to transform checkout flows into ESG-compliant, gamified experiences. Our 3D WebGL Earth visualization provides a tactile, emotional connection to environmental data, moving carbon tracking from "boring spreadsheets" to "magical interactivity."

WHAT I LEARNED

This project taught me the complexity of Spatial UI. Implementing mathematical shaders to reflect real-world data taught me how to bridge the gap between abstract metrics and visual storytelling. I also mastered the art of collaborative API design, ensuring that our headless engine could be integrated into any host architecture with just two lines of code.

Live Website ↗ YouTube ▶ GitHub Repository

4. CrisisGuard AI

Privacy-First Mental Health Crisis Detection using Edge Computing.

TYPE	ROLE
Self-Initiated Project	Individual Project

ONE-SENTENCE SUMMARY

A clinical-grade NLP system detecting 9 categories of mental health crises that runs entirely in the browser, ensuring absolute data privacy.

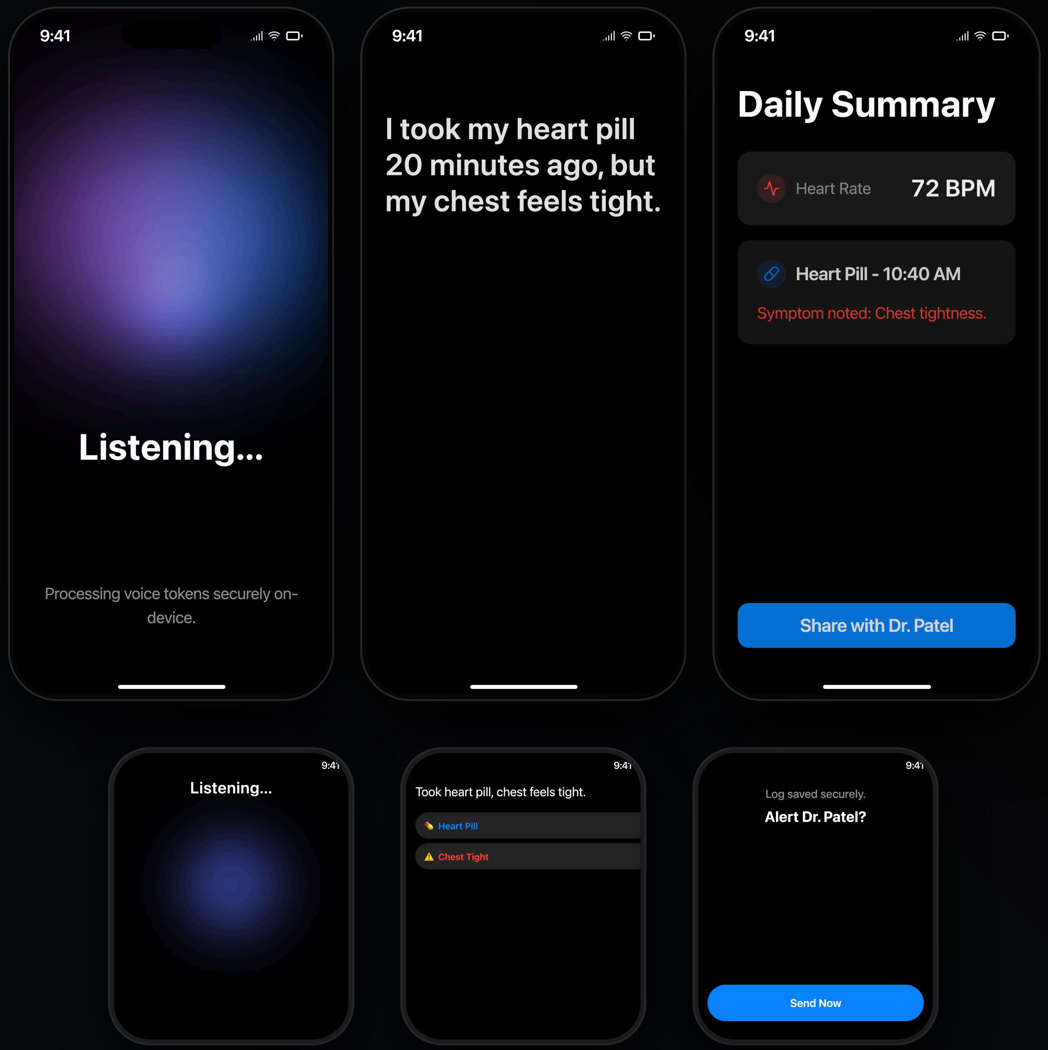
THE IMPACT

Analyzes deeply personal mental health text while guaranteeing **zero data surveillance**. It features a severity scoring system and a GPS-based counselor finder for 13+ countries. Successfully deployed live and selected for the UOE Summer of Code 2026.

WHAT I LEARNED

I learned that in healthcare, privacy is a prerequisite for trust. By mastering "Edge Computing", I figured out how to run complex Natural Language Processing models entirely client-side using JavaScript, avoiding centralized cloud processing.

Live Website ↗ YouTube ▶ GitHub



5. Vocalis — Voice-First Accessibility Concept for iOS & watchOS

TYPE	ROLE
Self-Initiated Concept Framework	Solo UX Architect & System Designer

ONE-SENTENCE SUMMARY

An on-device accessibility framework mapping complex medical NLP tokenization into a zero-touch, voice-driven interface.

THE IMPACT

Raw backend logic is useless if the interface lacks human empathy. Standard health applications actively exclude elderly or motor-impaired users through complex touchscreen demands. Applying strict multimedia design principles, I architected this concept to translate my existing web-based NLP engine into an Apple HIG-compliant, multimodal voice interface. By shifting the input method from screen-tapping to ambient voice logging via watchOS, the system restores digital healthcare access to vulnerable demographics without compromising local data privacy.

WHAT I LEARNED & MY OBJECTIVE

Mastering these constraints taught me the spatial psychology separating iOS interfaces from glanceable watchOS environments. I have the backend logic built and the visual architecture locked. My explicit goal for the Apple Developer Academy is to master native Swift and CoreML to compile this conceptual framework into deployed, wearable hardware.